Yes. Once we fix the dataset pipeline, **do not immediately jump to running all VLMs**. There should be a controlled sequence after that.

Your roadmap gives a very clear progression: full benchmark → prompt finalisation → model inference → automatic evaluation → human evaluation → statistical analysis → paper/release.

The next workflow

Step 1 — Finish and freeze the dataset

After fixing the issues we identified:

```
Remote sensing pipeline
        ↓
40–60 pilot
        ↓
quality validation
        ↓
label validation
        ↓
300–500 final patches
        ↓
spatial train/dev/test split
        ↓
FREEZE DATASET
```

At this point, **do not modify the test set anymore**.

Your final dataset should contain something like:

```
patch_001/
    before_rgb
    after_rgb
    before_falsecolor
    after_falsecolor
    before_ndvi
    after_ndvi
    before_evi
    after_evi
    before_vv
    after_vv
    before_vh
    after_vh

metadata.json
reference_label
```

The evidence card should contain the corresponding numerical evidence, including NDVI/EVI, VV/VH, temperature and rainfall anomalies, season and data-quality information.

Step 2 — Create the VLM evidence cards

This is where your project starts becoming a **VLM research project** rather than only a remote-sensing preprocessing project.

For each patch, create a standardized input.

For example:

```
PATCH ID: P042

BEFORE:
[RGB image]
[False-colour image]

AFTER:
[RGB image]
[False-colour image]

VEGETATION:
NDVI before: 0.72
NDVI after:  0.49
NDVI change: -0.23

EVI before: 0.61
EVI after:  0.43
EVI change: -0.18

SAR:
VV change: -1.8 dB
VH change: -2.7 dB

CLIMATE:
Temperature anomaly: +1.2°C
Rainfall anomaly: -65 mm

DATA QUALITY:
Cloud contamination: low
Season: January–February
```

Do not give coordinates or location names to the VLM, because the model could exploit geographic memorization rather than actually interpreting the evidence. Your protocol explicitly calls for this.

Step 3 — Freeze your experimental protocol

Before querying the models, decide:

Models

Your roadmap proposes:

1. One strong proprietary VLM
2. One additional accessible hosted VLM
3. One open-source VLM

For every model, record:

```
model name/version
checkpoint/API version
date accessed
temperature
image resolution
prompt version
token settings
number of repetitions
```

Do not change models halfway through the final experiment.

Step 4 — Define the experimental conditions

This is one of the most interesting parts of your project.

You don't simply ask:

"Which VLM is best?"

Instead, ask:

What happens when we give the VLM different types of evidence?

Condition A — Images only

```
Before RGB
After RGB
Before false colour
After false colour
```

Tests:

visual interpretation

Condition B — Indicators only

```
NDVI
EVI
VV
```

```
VH
temperature
rainfall
```

Tests:

numerical/tabular reasoning

Condition C — Combined

```
Images
+
Indicators
+
Climate
```

Tests:

multimodal evidence integration

Condition D — Structured explanation

Ask the model to separately provide:

```
supporting evidence
conflicting evidence
conclusion
uncertainty
```

This is optional in the roadmap but potentially very useful.

Step 5 — Design and freeze the prompt

This is **very important**.

You should not continuously modify your prompt after seeing test results.

Instead:

```
TRAIN/DEV
   ↓
try prompts
   ↓
compare
   ↓
select prompt
   ↓
PROMPT v1.0
   ↓
FREEZE
   ↓
```

```
TEST
```

Your roadmap specifically allocates Days 23–25 to prompt finalisation using development data, followed by a frozen experiment configuration.

Step 6 — Force structured VLM output

Don't ask:

"Explain what happened."

That makes evaluation difficult.

Instead require something like:

```
{
  "change_class": "vegetation_decline",
  "disturbance_present": true,
  "supporting_evidence": [
    "NDVI decreased substantially",
    "EVI also decreased",
    "VH decreased"
  ],
  "conflicting_evidence": [
    "No independent fire evidence"
  ],
  "explanation": "...",
  "confidence": 0.82,
  "insufficient_evidence": false
}
```

This follows the structured output proposed in your research plan.

And importantly, you ask for a **short evidence-based rationale**, not hidden chain-of-thought.

Step 7 — Run the VLM experiment

Now the actual experiment begins.

Suppose:

```
400 patches
×
3 models
×
3 conditions
```

That's:

3,600 model evaluations

before repetitions.

Every model should evaluate the **same patches** under the corresponding conditions.

For example:

	Model A	Model B	Model C
Patch 001			
Images only	✓	✓	✓
Indicators only	✓	✓	✓
Combined	✓	✓	✓
Patch 002			
Images only	✓	✓	✓
Indicators only	✓	✓	✓
Combined	✓	✓	✓
...			

This repeated-measures structure is important because you are comparing models and conditions on the **same underlying patches**.

Step 8 — Save everything

Do **not** save only the final answer.

For every inference, preserve:

```
patch_id
model
model_version
condition
prompt_version
temperature
timestamp
raw_response
parsed_response
classification
confidence
supporting_evidence
conflicting_evidence
error/failure
```

Your roadmap explicitly says to preserve raw outputs, metadata, failures, usage and cost.

This is essential for reproducibility.

Step 9 — Automatic evaluation

Now we compare VLM predictions against your **frozen reference labels**.

Classification

Calculate:

```
Accuracy
Balanced Accuracy
Macro Precision
Macro Recall
Macro-F1
Per-class F1
Confusion Matrix
```

Your **primary classification metric is Macro-F1**.

Step 10 — Evaluate the explanations

This is actually the **core contribution** of your project.

You ask:

Did the model's explanation actually use the evidence we supplied?

For example:

Model says:

"The forest declined because rainfall decreased."

But your evidence only shows:

```
Rainfall anomaly = -20 mm
NDVI = stable
EVI = stable
```

That's potentially:

```
unsupported causal claim
```

So evaluate:

```
Evidence precision
Evidence recall
Evidence F1
Unsupported-evidence rate
```

```
Contradiction rate
Evidence omission
Evidence correctness
Evidence completeness
Causal restraint
```

These are explicitly part of your planned evaluation.

Step 11 — Confidence calibration

Suppose the model says:

```
confidence = 0.95
```

but is correct only 65% of the time.

That's **overconfidence**.

So evaluate:

```
Brier score
Expected Calibration Error
Reliability diagrams
Accuracy by confidence interval
```

Also measure:

```
confidence variance
class agreement across repeated runs
explanation consistency
```

Again, these are already specified in your protocol.

Step 12 — Human evaluation

Now humans evaluate a subset of the explanations.

Your roadmap proposes:

```
80–100 explanations
        ↓
blind + randomize
        ↓
human ratings
```

They can evaluate things such as:

```
Is the explanation correct?
Is the evidence actually used?
Is it complete?
Is it useful?
Does it make unsupported claims?
```

The key is:

Humans should not know which model produced the explanation.

This reduces evaluator bias.

Step 13 — Statistical analysis

Now you have actual experimental results.

Because the same patch is evaluated under multiple models/conditions, this is a **repeated-measures design**.

Your planned analysis includes:

```
Bootstrap confidence intervals
        ↓
Paired comparisons
        ↓
Mixed-effects logistic regression
        ↓
Ordinal mixed-effects models for human ratings
        ↓
Holm correction
        ↓
Effect sizes + p-values
```

with:

```
Model + condition → fixed effects
Patch → random effect
```

Step 14 — Test your hypotheses

Then you answer your five hypotheses.

H1

Does combined evidence improve classification over images alone?

```
Images only
     vs
```

```
Combined
```

H2

Does multimodal evidence improve evidence coverage but potentially increase unsupported causal claims?

H3

Do models differ in classification, groundedness and calibration?

H4

Are models overconfident?

H5

Do structured explanations reduce contradictions?

These hypotheses are already pre-registered in your roadmap.

Step 15 — Write the paper

Your final paper is then roughly:

```
1. Introduction
        ↓
2. Related Work
        ↓
3. Dataset / Benchmark Construction
        ↓
4. Experimental Protocol
        ↓
5. VLMs
        ↓
6. Evaluation Metrics
        ↓
7. Results
        ↓
8. Statistical Analysis
        ↓
9. Error Analysis
        ↓
10. Limitations
        ↓
11. Conclusion
```

The contribution is **not**:

"We built a better forest detection model."

Your contribution is:

A reproducible benchmark and controlled evaluation of VLMs for forest-change classification and evidence-grounded ecological explanations.

So your immediate next step

I would **not start VLM inference yet**.

We should do this:

Phase 1 — Fix notebook

```
✓ EVI
✓ ERA5 monthly anomaly
✓ rainfall anomaly
✓ SAR handling
✓ fire evidence
✓ independent labeling
✓ spatial split
✓ alignment QA
✓ human validation
✓ final metadata
```

Phase 2 — Run only 40–60 samples

Then verify:

```
✓ Every image is correct
✓ Every label is defensible
✓ Every evidence card is correct
✓ JSON output works
✓ Prompt works
✓ VLM doesn't get location leakage
✓ Evaluation scripts work
```

This is exactly why the roadmap says to **pilot first and scale only after data-quality and prompt-validation checks pass**.

Phase 3 — Freeze everything

```
DATASET v1.0
PROMPT v1.0
MODELS v1.0
PROTOCOL v1.0
```

Phase 4 — 300–500 final benchmark

Phase 5 — VLM inference

Phase 6 — Explanation evaluation

Phase 7 — Human evaluation

Phase 8 — Statistical analysis

Phase 9 — Paper

In one sentence:

After fixing the current notebook, our next real task is to build and validate the 40–60-sample pilot + finalize the VLM evidence-card format and prompt; only after that should we scale to the 300–500-sample benchmark and run the controlled VLM experiment.